Solar Panels

Pupil sheet

Name: _____ **Date:** _____

Safety: Never look at the sun: find the angle from the shadow. The box is not waterproof: keep it off wet grass.

1. What I see on the screen

The screen shows three cards: **Illuminance** (lx), **Irradiance (model)** (W/m^2) and **Estimated power** (W). **Only the illuminance is measured.** The other two are worked out from it with a model, $E = L/120$ and $P = \eta A E$; they would become measurements only with a real panel and a multimeter.

Wait 20 **seconds** after every move, long enough for your hand to leave the field. **No shadow over the sensor** — not yours and not the prop's. Temperature takes longer to settle: 60–90 seconds.

2. What I do on my own

First I predict. I write it and date it, and I change nothing after I measure.

	predict	measure
(a) How many times brighter is the sun than the room?	_____	_____
(b) Tilted by 45°, how much power is left?	_____	_____
(c) Better on a hot day or a cool bright one?	_____	_____

Panel parameters (from the parameter step): area _____ m^2, efficiency _____ %

Three places. The last column is the ratio to the room value — that is the result worth remembering, not the raw number.

Place	Light (lx)	Est. power (W)	Times the room
In the room			1
Beside the window			
In the sun			

The tilt series. $\cos\theta$ is already filled in. I find 0° by the **smallest shadow** of the board — not by looking at the sun. I choose the angle in the list before **Capture**; the last one is **0° again (control)**. The whole series is done in one go, under an unchanged sky.

Angle θ	$\cos\theta$	**Light (lx)**	**Ratio** $L(\theta)/L(0)$
0°	1.00		1.00
15°	0.97		
30°	0.87		
45°	0.71		
60°	0.50		
75°	0.26		
90°	0.00		
0° (control)	1.00		

Sky (circle one): clear / broken / overcast **Time:** _____

The temperature correction. $T_{\text{air}} =$ _____ °C, $E =$ _____ W/m^2, so $T_{\text{panel}} = T_{\text{air}} + 0.031\,E =$ _____ °C and $\eta = \eta_0\,[1 - 0.004\,(T_{\text{panel}} - 25)] =$ _____ % *(below 25 °C it comes out slightly above η_0).* I check it on the **Corrected efficiency** card.

3. The conclusion I write

I predicted that, and the data show that _____

_____

At 90° I still measured something, because _____

_____

The power on the screen is estimated, not measured, because _____

_____

Exit ticket

Which of the numbers on the screen were measured and which were worked out?

_____